

THE HUMP BETWEEN TWO CLOSE ZEROS: AN EXACT SMALL-GAP LAW FOR RANDOM UNITARY MATRICES AND A TILT LADDER FOR THE RIEMANN ZETA FUNCTION

UĞUR SEZEN

ABSTRACT. Let M be the maximum of $|\Lambda_N|$, the characteristic polynomial of a Haar unitary $N \times N$ matrix, on the arc between two consecutive eigenangles at distance s . For fixed N we prove that, conditionally on $s < \varepsilon$, the pair $(s/\varepsilon, M/s^2)$ converges in law as $\varepsilon \rightarrow 0$ to independent variables $(U^{1/3}, X_{N-2}/4)$, with U uniform and X_{N-2} the value of $|\Lambda_{N-2}(1)|$ under the $|\Lambda_{N-2}(1)|^4$ -tilted Haar measure; the mean number of such gaps is $N^2(N^2 - 1)\varepsilon^3/(72\pi)$ to leading order, and all moments of order $2k > -5$ and all log-moments converge. Summed over all gaps, $\mathbb{E} \sum_n M_n^2 \sim \frac{\varepsilon^2-5}{2}N^2$, by an identity that reduces it to a joint moment evaluated by Winn. By the product formula of Bourgade, Hughes, Nikeghbali and Yor, X_{N-2} is a product of independent factors, so moments and log-cumulants are explicit: in unfolded units the log-hump has mean $\rightarrow 2\log \pi + 2\gamma - 10/3$, variance $\frac{1}{2}\log N + O(1)$ and a negative third cumulant ($\rightarrow -0.1271$), and is asymptotically normal. For the Riemann zeros we cast the analogue as a tilt ladder (conditioning on $b = 0, 1, 2$ zeros) and report pre-registered tests at $L = \log(t/2\pi) \approx 9-22$: the arithmetic shift of the log-variance is close to the Keating–Snaith value -0.088 on every rung; the shift of the third cumulant falls from $+0.28$ to $+0.15$ and $+0.086$ at $b = 2$ (predicted blind within 1.6σ) and flattens with height, as a finite-height correction would. In a further test, single zeros follow Landau’s formula for the phases of the prime waves to four decimals, and the midpoints of close pairs lock to these phases 3.87 ± 0.02 times as strongly, close to the factor 4 of a local-GUE heuristic.

1. INTRODUCTION

Let $\gamma_n < \gamma_{n+1}$ be consecutive ordinates of zeros of the Riemann zeta function on the critical line, $\delta_n = \gamma_{n+1} - \gamma_n$, and let $M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$ be the height of Hardy’s function between them. The mean of M_n^2 over all gaps is known [11, 18]; the *joint* law of (δ_n, M_n) was studied numerically in [1], whose first open problem asked for the mean of M_n^2 conditioned on the gap. A reader asked the sharper question: in the regime of asymptotically small gaps, can the joint distribution of (δ_n, M_n^2) be determined?

For random unitary matrices the answer is yes, and exact. When two zeros nearly coincide, the hump between them is crushed by the square of the gap, and what is left is the height produced at the collision point by all the *other* zeros. For the characteristic polynomial of a Haar unitary matrix, conditioning two eigenvalues to merge at a point tilts the remaining $N - 2$ eigenvalues by $|\Lambda_{N-2}|^4$ at that point; under this tilt the value $|\Lambda_{N-2}|$ is a product of independent factors, so every moment and every cumulant of the logarithm is explicit.

Results for the unitary group. Let s be a gap between consecutive eigenangles of $U \in U(N)$ and M the maximum of $|\Lambda_N|$ on it (Section 2).

- *Theorem 1* (fixed N , $\varepsilon \rightarrow 0$). The expected number of gaps below ε is $N^2(N^2 - 1)\varepsilon^3/(72\pi)(1 + o(1))$; conditionally on $s < \varepsilon$ (for a gap chosen uniformly),

$(s/\varepsilon, M/s^2) \Rightarrow (U^{1/3}, X_{N-2}/4)$ with U uniform on $[0, 1]$ and *independent* of X_{N-2} , the value $|\Lambda_{N-2}(1)|$ under the $|\Lambda_{N-2}(1)|^4$ -tilted Haar measure. All moments $\mathbb{E}(M/s^2)^{2k}$ with $k > -5/2$ and all log-moments converge; the threshold is sharp.

- *Theorem 2.* X_n has the law of $\prod_{j=1}^n |1 - \xi_j|$ with independent, explicitly distributed ξ_j ; $\mathbb{E}X_n^{2k}$ is a ratio of Keating–Snaith moments and $\kappa_r(\log X_n) = \sum_{j=1}^n [\psi^{(r-1)}(j+4) - 2^{1-r}\psi^{(r-1)}(j+2)]$.
- *Corollary 3* (large N , unfolded units). The moments grow like N^{k^2} with the constant $\pi^{4k} 12 G(k+3)^2 / G(2k+5)$; the log-hump has mean $\rightarrow 2 \log \pi + 2\gamma - 10/3$, variance $\frac{1}{2} \log N + c_2 + o(1)$ with $c_2 = 65/18 + \gamma/2 - \pi^2/2$, third cumulant negative for every N and tending to -0.127077 , and is asymptotically Gaussian.
- *Corollary 4 and Proposition 5.* The sum of M_n^2 over the gaps below ε has mean $\mathcal{M}_{N-2}(3) \varepsilon^7 / (224\pi)$ to leading order, while the sum over all gaps has mean $2\pi \mathbb{E}|V_N(0)V_N'(0)| \sim \frac{e^2-5}{2} N^2$ (V_N the real form of Λ_N , the constant from Winn [26]): gaps below $\tilde{\varepsilon}$ in unfolded units carry a share of order $\tilde{\varepsilon}^7$ of the second moment.

What is known. To our knowledge, the joint (gap, hump) law has not been written down before, but its ingredients are known. That conditioning the eigenvalues on points at a given location tilts the remaining points by powers of the characteristic polynomial is the Palm-measure structure of these ensembles (for the unitary group it is immediate from Weyl’s formula; see [14] for the log-gas setting). The Haar measure tilted by $|\det(1 - U)|^{2\delta}$ is the circular Jacobi ensemble [14, 9], which for $\delta \in \mathbb{N}$ is the CUE conditioned to have δ further eigenvalues at 1 [9, §1.1], and the product formula for $\det(1 - U)$ under Haar and tilted measures is due to Bourgade, Hughes, Nikeghbali and Yor [8] and Bourgade, Nikeghbali and Rouault [9]. Theorem 2 is a direct consequence of their results and is presented as such. The small gaps of the CUE were determined by Ben Arous and Bourgade [6] (Poisson limit at scale $N^{-4/3}$), and small gaps of other ensembles in [12, 13, 7]; these works concern the gaps, not the polynomial between the eigenvalues. Theorem 1(a) agrees with the Ben Arous–Bourgade intensity. The proof of Theorem 1 is elementary (Weyl’s formula and dominated convergence); we give it in full because the negative moments require a uniform domination, which turns out to follow from a one-line geometric inequality (Lemma B). For the total second moment we use the random-matrix counterpart of the reduction of Conrey and Ghosh (sum of squared maxima = integral of $|VV'|$) and Winn’s evaluation of the resulting joint moment [26].

The zeta side. Conditioning on b zeros at a point tilts the rest of the spectrum by $|\Lambda|^{2b}$: $b = 0$ is the value distribution of $\log |\zeta|$ (Keating–Snaith [19]), $b = 1$ is $\log |\zeta'(\rho)|$ (Hughes–Keating–O’Connell [17]; see [10]), and $b = 2$ is the log-hump of close pairs studied here. Adding the Keating–Snaith arithmetic factor a_k to each rung gives a parameter-free prediction for the cumulants (Section 8). We tested it in three pre-registered steps on Riemann zeros (Section 9): the arithmetic shift of the variance is rung-independent and close to a_k , while the arithmetic shift of the third cumulant decreases from rung to rung at $L \approx 9$ –12 — a decrease that was predicted blind for $b = 2$ from the two lower rungs — and the ladder flattens at larger heights. A fourth pre-registered test looks at the phases of the prime waves at the conditioned points: single zeros obey Landau’s formula to four decimals, and midpoints of close pairs lock about four times as strongly, as a local-GUE heuristic predicts; this lock has the predicted size and does not by itself explain the decay along the ladder, which remains open. We report the failed parts of the predictions next to the successful ones. Nothing here bears on a proof of the Riemann Hypothesis.

2. SETUP AND NOTATION

Eigenangles. Let U be Haar distributed on $U(N)$, $N \geq 2$, with eigenvalues $e^{i\theta_1}, \dots, e^{i\theta_N}$, the labelled eigenangles $\theta = (\theta_1, \dots, \theta_N) \in \mathbb{T}^N$, $\mathbb{T} = [0, 2\pi)$, being exchangeable with density (Weyl's integration formula)

$$(1) \quad p_N(\theta) = \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2.$$

The density is invariant under the rotation $\theta_j \mapsto \theta_j + \alpha$; almost surely the θ_j are distinct. For $n \geq 0$ we write $\mathbb{E}_n$ for the expectation under p_n ($\mathbb{E}_0$ is trivial) and

$$|\Lambda_n(e^{i\vartheta})| = \prod_{j=1}^n |e^{i\vartheta} - e^{i\theta_j}| = \prod_{j=1}^n 2 \left| \sin \frac{\vartheta - \theta_j}{2} \right|, \quad \Lambda_0 \equiv 1,$$

for the modulus of the characteristic polynomial $\det(e^{i\vartheta} - U)$; only the modulus is used.

Keating–Snaith moments. For $n \geq 1$ and real k ,

$$(2) \quad \mathcal{M}_n(k) := \mathbb{E}_n |\Lambda_n(1)|^{2k} = \prod_{j=1}^n \frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2} \quad (k > -\tfrac{1}{2}),$$

and $\mathcal{M}_n(k) = \infty$ for $k \leq -\frac{1}{2}$ [19] (see also the proof of Theorem 1(c)); $\mathcal{M}_0 \equiv 1$. By rotation invariance $\mathbb{E}_n |\Lambda_n(e^{i\alpha})|^{2k} = \mathcal{M}_n(k)$ for every α . Telescoping the products gives

$$(3) \quad \begin{aligned} \mathcal{M}_n(2) &= \frac{(n+1)(n+2)^2(n+3)}{12}, \\ \mathcal{M}_n(3) &= \frac{(n+1)(n+2)^2(n+3)^3(n+4)^2(n+5)}{8640}. \end{aligned}$$

Gaps and humps. Order the eigenangles, $\theta_{(1)} < \dots < \theta_{(N)}$, put $\theta_{(N+1)} = \theta_{(1)} + 2\pi$, and let

$$s_n = \theta_{(n+1)} - \theta_{(n)}, \quad M_n = \max_{\theta_{(n)} \leq \vartheta \leq \theta_{(n+1)}} |\Lambda_N(e^{i\vartheta})|, \quad n = 1, \dots, N.$$

The unfolded gap is $\tilde{s}_n = N s_n / 2\pi$ (mean one). For a function G_n of the n -th gap we write

$$(4) \quad \langle G \rangle_\varepsilon := \frac{\mathbb{E} \sum_{n=1}^N \mathbf{1}\{s_n < \varepsilon\} G_n}{\mathbb{E} \#\{n : s_n < \varepsilon\}} = \mathbb{E}[G_J \mid s_J < \varepsilon],$$

where J is uniform on $\{1, \dots, N\}$ and independent of U : this is the law of a typical gap conditioned to be small.

The tilted law. For $n \geq 1$ let $\tilde{\mathbb{P}}_n$ be the Haar measure of $U(n)$ tilted by $|\Lambda_n(1)|^4 / \mathcal{M}_n(2)$, $\tilde{\mathbb{E}}$ its expectation, and X_n the random variable $|\Lambda_n(1)|$ under $\tilde{\mathbb{P}}_n$. This is the circular Jacobi ensemble of [9, Def. 1.1] with $\beta = 2$ and $\delta = 2$: their weight $(1 - e^{-i\theta_j})^\delta (1 - e^{i\theta_j})^\delta$ equals $|1 - e^{i\theta_j}|^{2\delta}$ for real δ . Thus

$$\tilde{\mathbb{E}} h(X_n) = \frac{\mathbb{E}_n [|\Lambda_n(1)|^4 h(|\Lambda_n(1)|)]}{\mathcal{M}_n(2)}.$$

We set $X_0 \equiv 1$. In unfolded units the limit object is $Y_N := (2\pi/N)^2 X_{N-2}/4$.

Special functions. $H_m^{(r)} = \sum_{i=1}^m i^{-r}$, $H_m = H_m^{(1)}$; $\psi = \Gamma'/\Gamma$ and $\psi^{(r)}$ its derivatives; γ is Euler's constant; G is the Barnes function, $G(z+1) = \Gamma(z)G(z)$, $G(1) = 1$; $\text{sinc } x = \sin x/x$. Cumulants are κ_r .

3. THE SMALL-GAP LIMIT

Theorem 1. *Let $N \geq 2$ and $B_N := 2^{N-2}/4$.*

- (i) *For every configuration of distinct eigenangles and every n , $0 < M_n/s_n^2 \leq B_N$.*
- (ii) *Let $F : (0, 1] \times (0, B_N] \rightarrow \mathbb{R}$ be continuous, and suppose that $|F(u, y)| \leq C(1 + y^{2k_0})$ for some $C < \infty$ and some $k_0 \in (-\frac{5}{2}, 0]$. Then*

(5)

$$\lim_{\varepsilon \downarrow 0} \varepsilon^{-3} \mathbb{E} \sum_{n=1}^N \mathbf{1}_{\{s_n < \varepsilon\}} F\left(\frac{s_n}{\varepsilon}, \frac{M_n}{s_n^2}\right) = \frac{1}{2\pi} \int_0^1 u^2 \mathbb{E}_{N-2} \left[|\Lambda_{N-2}(1)|^4 F\left(u, \frac{|\Lambda_{N-2}(1)|}{4}\right) \right] du.$$

In particular:

- (a) $\mathbb{E} \# \{n : s_n < \varepsilon\} = \frac{N^2(N^2 - 1)}{72\pi} \varepsilon^3 (1 + o(1)).$
- (b) *Under $\langle \cdot \rangle_\varepsilon$, $(s_J/\varepsilon, M_J/s_J^2) \Rightarrow (U^{1/3}, X_{N-2}/4)$ as $\varepsilon \rightarrow 0$, where U is uniform on $[0, 1]$ and independent of X_{N-2} . In the small-gap limit, gap and normalised hump are independent.*
- (c) *For every $k > -\frac{5}{2}$, $\langle (M/s^2)^{2k} \rangle_\varepsilon \rightarrow 4^{-2k} \widetilde{\mathbb{E}} X_{N-2}^{2k} = 4^{-2k} \mathcal{M}_{N-2}(k+2)/\mathcal{M}_{N-2}(2)$, and for every $m > 0$, $\langle |\log(M/s^2)|^m \rangle_\varepsilon \rightarrow \widetilde{\mathbb{E}} |\log(X_{N-2}/4)|^m$; the same holds for integer powers of $\log(M/s^2)$, so all cumulants of $\log(M/s^2)$ converge. If $N \geq 3$ and $k \leq -\frac{5}{2}$, then $\langle (M/s^2)^{2k} \rangle_\varepsilon \rightarrow \infty$.*

The factor $|\Lambda_{N-2}(1)|^4$ in (5) is the product of the two factors $|\Lambda_{N-2}(e^{i\theta})|^2$ that the Vandermonde determinant attaches to each of the two merging eigenvalues; u^2 is their mutual repulsion. Ben Arous and Bourgade [6, Thm. 1.1] showed that the points $(N^{4/3}(\theta_{(n+1)} - \theta_{(n)}), \theta_{(n)})$ (gaps in angle units) converge to a Poisson process with intensity $\frac{1}{24\pi} u^2 du \frac{d\theta}{2\pi}$, so that the mean number of gaps below $xN^{-4/3}$ tends to $\int_0^x u^2 du / (24\pi) = x^3/(72\pi)$. Setting $\varepsilon = xN^{-4/3}$ in (a) gives the same leading term, $N^2(N^2 - 1)x^3N^{-4}/(72\pi) \rightarrow x^3/(72\pi)$. The two limits are taken in different orders ($\varepsilon \rightarrow 0$ at fixed N here, $N \rightarrow \infty$ at fixed x there); the constants agree.

3.1. Two deterministic lemmas. Fix $s \in (0, 2\pi)$ and a configuration $\vartheta = (\vartheta_1, \dots, \vartheta_{N-2}) \in \mathbb{T}^{N-2}$ of the remaining angles, with $|\Lambda_{N-2}|$ built from ϑ . Put

$$M(s; \vartheta) := \max_{0 \leq t \leq s} |e^{it} - 1| |e^{it} - e^{is}| |\Lambda_{N-2}(e^{it})|,$$

$$W_s(\vartheta) := \prod_{j=1}^{N-2} |e^{i\vartheta_j} - 1|^2 |e^{i\vartheta_j} - e^{is}|^2,$$

$$A_s := \{\vartheta : \vartheta_j \notin (0, s) \text{ for all } j\}.$$

$M(s; \vartheta)$ is the hump over the arc $[0, s]$ of the polynomial with zeros at 1, e^{is} and $e^{i\vartheta_j}$; W_s is the Vandermonde coupling between the pair and the rest; A_s says that the arc is a gap. For $N = 2$ all products over j are empty.

Lemma A (Hump bounds). *For every $s \in (0, 2\pi)$ and every ϑ ,*

$$(6) \quad \frac{4 \sin^2(s/4)}{s^2} |\Lambda_{N-2}(e^{is/2})| \leq \frac{M(s; \vartheta)}{s^2} \leq \frac{1}{4} \max_{0 \leq t \leq s} |\Lambda_{N-2}(e^{it})| \leq B_N.$$

Consequently $M(s; \vartheta)/s^2 \rightarrow |\Lambda_{N-2}(1)|/4$ as $s \downarrow 0$, for every ϑ ; and $M(s; \vartheta)/s^2 \geq \frac{1}{5} |\Lambda_{N-2}(e^{is/2})|$ for $0 < s \leq 1$.

Proof. For $t \in [0, s]$ the numbers $t/2$ and $(s-t)/2$ lie in $[0, \pi]$, so $|e^{it} - 1| = 2 \sin(t/2)$ and $|e^{it} - e^{is}| = 2 \sin((s-t)/2)$, and

$$q_s(t) := 4 \sin \frac{t}{2} \sin \frac{s-t}{2} = 2 \left[\cos \left(t - \frac{s}{2} \right) - \cos \frac{s}{2} \right].$$

Thus q_s is maximal at $t = s/2$, where it equals $2(1 - \cos \frac{s}{2}) = 4 \sin^2 \frac{s}{4}$; evaluating the product at $t = s/2$ gives the lower bound. Since $0 \leq \sin x \leq x$ for $x \in [0, \pi]$, $q_s(t) \leq t(s-t) \leq s^2/4$, which gives the middle bound, and $|e^{it} - e^{is}| \leq 2$ gives the last one. As $s \downarrow 0$, $4 \sin^2(s/4)/s^2 = \frac{1}{4} \text{sinc}^2(s/4) \rightarrow \frac{1}{4}$ and both $|\Lambda_{N-2}(e^{is/2})|$ and $\max_{[0, s]} |\Lambda_{N-2}|$ tend to $|\Lambda_{N-2}(1)|$ by continuity. Finally, sinc is decreasing on $(0, \pi)$, so for $s \leq 1$, $4 \sin^2(s/4)/s^2 \geq 4 \sin^2(1/4) = 0.2448 \dots > \frac{1}{5}$. $\square$

Part (i) of Theorem 1 follows: after a rotation, the n -th gap is the arc $[0, s_n]$ with ϑ the other eigenangles, so $M_n = M(s_n; \vartheta)$; the upper bound is (6), and $M_n > 0$ because $M_n \geq |\Lambda_N|$ at the midpoint of the gap, which is not an eigenangle.

Lemma B (Domination). *For every $s \in (0, 2\pi)$ and every $\vartheta \in A_s$,*

$$(7) \quad W_s(\vartheta) \leq |\Lambda_{N-2}(e^{is/2})|^4.$$

Consequently, for $k \in (-\frac{5}{2}, 0]$, $0 < s \leq 1$ and $\vartheta \in A_s$,

$$(8) \quad W_s(\vartheta) \left(\frac{M(s; \vartheta)}{s^2} \right)^{2k} \leq 5^{-2k} |\Lambda_{N-2}(e^{is/2})|^{4+2k},$$

and $\mathbb{E}_{N-2} |\Lambda_{N-2}(e^{is/2})|^{4+2k} = \mathcal{M}_{N-2}(k+2) < \infty$.

The inequality (7) is index by index: for a point outside an arc, the product of its distances to the two endpoints of the arc is at most the square of its distance to the midpoint.

Proof. Let $\vartheta_j \notin (0, s)$ and let $x_j \in (-\pi, \pi]$ be the representative of $\vartheta_j - s/2$ modulo 2π . The points of $(0, s)$ are exactly those at circular distance less than $s/2$ from $s/2$, so $s/2 \leq |x_j| \leq \pi$. Since $|\sin(y/2)|$ is 2π -periodic,

$$|e^{i\vartheta_j} - 1| |e^{i\vartheta_j} - e^{is}| = 4 \left| \sin \frac{x_j + s/2}{2} \sin \frac{x_j - s/2}{2} \right| = 2 \left| \cos \frac{s}{2} - \cos x_j \right|,$$

by $2 \sin a \sin b = \cos(a-b) - \cos(a+b)$. Now $\cos \frac{s}{2} - \cos x_j = 2 \sin^2 \frac{x_j}{2} - 2 \sin^2 \frac{s}{4}$, and since $s/4 \leq |x_j|/2 \leq \pi/2$ and $\sin^2$ is increasing on $[0, \pi/2]$, $0 \leq \cos \frac{s}{2} - \cos x_j \leq 2 \sin^2 \frac{x_j}{2}$. Hence

$$|e^{i\vartheta_j} - 1| |e^{i\vartheta_j} - e^{is}| \leq 4 \sin^2 \frac{x_j}{2} = |e^{i\vartheta_j} - e^{is/2}|^2,$$

and squaring and multiplying over j gives (7). For (8), combine (7) with the lower bound $M/s^2 \geq \frac{1}{5} |\Lambda_{N-2}(e^{is/2})|$ of Lemma A (raised to the power $2k \leq 0$). The expectation equals $\mathcal{M}_{N-2}(k+2)$ by rotation invariance and is finite because $k+2 > -\frac{1}{2}$. $\square$

Remark 1 (Several angles near the midpoint). The bound (8) does not require the singular factors to be treated one cluster at a time: configurations in which several ϑ_j approach the midpoint simultaneously are included in the exact moment $\mathcal{M}_{N-2}(k+2)$. For a self-contained check of its finiteness, bound the Vandermonde factor of p_{N-2} by $4^{(N-2)(N-3)/2}$; then by Tonelli $\mathbb{E}_{N-2} |\Lambda_{N-2}(1)|^{2a} \leq c_N \prod_{j=1}^{N-2} \int_{\mathbb{T}} |2 \sin(\vartheta_j/2)|^{2a} d\vartheta_j$, a product of one-dimensional integrals, each finite iff $2a > -1$. With $2a = 4 + 2k$ this is $k > -\frac{5}{2}$.

Remark 2 (Uniform integrability). Lemma B and Lemma C below give, for $k \in (-\frac{5}{2}, 0]$ and $0 < \varepsilon \leq 1$,

$$\varepsilon^{-3} \mathbb{E} \sum_n \mathbf{1}\{s_n < \varepsilon\} \left(\frac{M_n}{s_n^2} \right)^{2k} \leq \frac{5^{-2k} \mathcal{M}_{N-2}(k+2)}{6\pi},$$

uniformly in ε . Together with (a) this bounds $\sup_{\varepsilon \leq \varepsilon_0} \langle (M/s^2)^{2k} \rangle_\varepsilon$, so by the de la Vallée-Poussin criterion $(M_J/s_J^2)^{2k}$ is uniformly integrable under $\langle \cdot \rangle_\varepsilon$ for every $k > -\frac{5}{2}$ (use an exponent $k' \in (-\frac{5}{2}, k)$). The proof below uses the domination directly.

3.2. Reduction to a two-point integral.

Lemma C. *Let $N \geq 2$, $0 < \varepsilon < 2\pi$, and let $g : (0, 2\pi) \times (0, \infty) \rightarrow [0, \infty]$ be measurable. Then*

(9)

$$\varepsilon^{-3} \mathbb{E} \sum_{n=1}^N \mathbf{1}\{s_n < \varepsilon\} g(s_n, M_n) = \frac{1}{2\pi} \int_0^1 u^2 \operatorname{sinc}^2\left(\frac{\varepsilon u}{2}\right) \mathbb{E}_{N-2} \left[W_{\varepsilon u} \mathbf{1}_{A_{\varepsilon u}} g(\varepsilon u, M(\varepsilon u; \cdot)) \right] du,$$

where $\mathbb{E}_{N-2}$ integrates over ϑ with density p_{N-2} . The same identity holds for real g whenever the right-hand side is finite with $|g|$ in place of g .

Proof. For distinct angles and an ordered pair (a, b) , $a \neq b$, let $\sigma_{ab} \in (0, 2\pi)$ be $\theta_b - \theta_a$ modulo 2π . Call b the successor of a if no θ_j lies on the open counterclockwise arc from θ_a to θ_b . The gaps are in bijection with the ordered pairs (a, b) such that b is the successor of a , the gap being σ_{ab} and the hump the maximum of $|\Lambda_N|$ over the closed arc. By exchangeability each of the $N(N-1)$ ordered pairs contributes equally, so the left side of (9) equals

$$\varepsilon^{-3} N(N-1) \int_{\mathbb{T}^N} p_N(\theta) \mathbf{1}\{\sigma_{12} < \varepsilon\} \mathbf{1}\{2 \text{ is the successor of } 1\} g(\sigma_{12}, M_{12}) d\theta.$$

Change variables $\theta_1 = \varphi$, $\theta_2 = \varphi + \varepsilon u \pmod{2\pi}$, $u \in (0, 1)$, which maps $\mathbb{T} \times (0, 1)$ bijectively onto $\{(\theta_1, \theta_2) : \sigma_{12} < \varepsilon\}$ with $d\theta_1 d\theta_2 = \varepsilon d\varphi du$, and put $\vartheta_j = \theta_{j+2} - \varphi \pmod{2\pi}$, a measure-preserving bijection of $\mathbb{T}^{N-2}$. The Vandermonde product splits into the pair factor $|1 - e^{i\varepsilon u}|^2 = 4 \sin^2(\varepsilon u/2) = \varepsilon^2 u^2 \operatorname{sinc}^2(\varepsilon u/2)$, the coupling $W_{\varepsilon u}(\vartheta)$, and the Vandermonde product of ϑ ; the successor condition becomes $\vartheta \in A_{\varepsilon u}$; and $M_{12} = M(\varepsilon u; \vartheta)$ because $|\Lambda_N(e^{i(\varphi+t)})| = |e^{it} - 1| |e^{it} - e^{i\varepsilon u}| |\Lambda_{N-2}(e^{it})|$. Nothing depends on φ , whose integral gives 2π . Finally

$$\frac{N(N-1)}{(2\pi)^N N!} \prod_{j < k} |e^{i\vartheta_j} - e^{i\vartheta_k}|^2 = \frac{1}{(2\pi)^2} p_{N-2}(\vartheta),$$

and collecting $\varepsilon^{-3} \cdot \varepsilon \cdot \varepsilon^2 \cdot 2\pi / (2\pi)^2 = 1/(2\pi)$ gives (9). The statement for real g follows by splitting $g = g^+ - g^-$. $\square$

3.3. Proof of Theorem 1. We use the following form of dominated convergence.

Lemma D. *On a measure space, let $|f_\varepsilon| \leq g_\varepsilon$, $f_\varepsilon \rightarrow f$ and $g_\varepsilon \rightarrow g$ almost everywhere, and $\int g_\varepsilon \rightarrow \int g < \infty$ as $\varepsilon \rightarrow 0$. Then $\int f_\varepsilon \rightarrow \int f$.*

Proof. A limit as $\varepsilon \rightarrow 0$ exists and equals $\int f$ if and only if it does so along every sequence $\varepsilon_n \rightarrow 0$, so it suffices to fix such a sequence; along it the hypotheses hold almost everywhere (the exceptional null set does not depend on the sequence) and Fatou's lemma applies. Applied to $g_{\varepsilon_n} + f_{\varepsilon_n} \geq 0$ and $g_{\varepsilon_n} - f_{\varepsilon_n} \geq 0$ it gives $\int g \pm \int f \leq \int g + \liminf_n (\pm \int f_{\varepsilon_n})$, i.e. $\limsup_n \int f_{\varepsilon_n} \leq \int f \leq \liminf_n \int f_{\varepsilon_n}$. $\square$

Proof of (ii). Apply Lemma C with $g(s, M) = F(s/\varepsilon, M/s^2)$ (admissible by part (i) and the bound below). For $0 < \varepsilon \leq 1$, $u \in (0, 1]$ and $\vartheta \in \mathbb{T}^{N-2}$ put $s = \varepsilon u$ and

$$f_\varepsilon(u, \vartheta) = u^2 \operatorname{sinc}^2\left(\frac{s}{2}\right) W_s(\vartheta) \mathbf{1}_{A_s}(\vartheta) F\left(u, \frac{M(s; \vartheta)}{s^2}\right),$$

so that the left side of (5), before the limit, is $(2\pi)^{-1} \int_0^1 \mathbb{E}_{N-2} f_\varepsilon(u, \cdot) du$.

Domination. On A_s , by the growth condition on F , $\operatorname{sinc} \leq 1$, and (7), (8) (with $k = k_0$),

$$|f_\varepsilon| \leq Cu^2 W_s \left(1 + \left(\frac{M}{s^2}\right)^{2k_0}\right) \leq Cu^2 \left(|\Lambda_{N-2}(e^{is/2})|^4 + 5^{-2k_0} |\Lambda_{N-2}(e^{is/2})|^{4+2k_0}\right) =: g_\varepsilon(u, \vartheta),$$

and $f_\varepsilon = 0$ off A_s . By rotation invariance $\int_0^1 \mathbb{E}_{N-2} g_\varepsilon du = \frac{C}{3} (\mathcal{M}_{N-2}(2) + 5^{-2k_0} \mathcal{M}_{N-2}(k_0 + 2)) < \infty$ for every ε .

Pointwise limits. Let ϑ be such that no ϑ_j is 0; this excludes a null set, and on its complement $\Lambda_{N-2}(1) \neq 0$. Fix $u \in (0, 1]$ and let $\varepsilon \rightarrow 0$. Then $\vartheta \in A_{\varepsilon u}$ as soon as $\varepsilon u < \min_j \vartheta_j$; $W_{\varepsilon u}(\vartheta) \rightarrow |\Lambda_{N-2}(1)|^4$ and $|\Lambda_{N-2}(e^{i\varepsilon u/2})| \rightarrow |\Lambda_{N-2}(1)| > 0$ by continuity; $M(\varepsilon u; \vartheta)/(\varepsilon u)^2 \rightarrow |\Lambda_{N-2}(1)|/4 \in (0, B_N]$ by Lemma A; and F is continuous at $(u, |\Lambda_{N-2}(1)|/4)$. Hence

$$\begin{aligned} f_\varepsilon &\rightarrow f_0 := u^2 |\Lambda_{N-2}(1)|^4 F(u, |\Lambda_{N-2}(1)|/4), \\ g_\varepsilon &\rightarrow g_0 := Cu^2 (|\Lambda_{N-2}(1)|^4 + 5^{-2k_0} |\Lambda_{N-2}(1)|^{4+2k_0}), \end{aligned}$$

and $\int_0^1 \mathbb{E}_{N-2} g_\varepsilon du = \int_0^1 \mathbb{E}_{N-2} g_0 du$ for all ε . Lemma D on $(0, 1] \times \mathbb{T}^{N-2}$ with the measure $du \otimes p_{N-2} d\vartheta$ gives (5). $\square$

Proof of (a)–(c). (a) Take $F \equiv 1$: the limit is $\frac{1}{2\pi} \cdot \frac{1}{3} \cdot \mathcal{M}_{N-2}(2) = N^2(N^2 - 1)/(72\pi)$ by (3).

(b) For bounded continuous F on $[0, 1] \times [0, \infty)$, divide (5) by (a):

$$\langle F(s/\varepsilon, M/s^2) \rangle_\varepsilon \rightarrow \int_0^1 3u^2 \tilde{\mathbb{E}}[F(u, X_{N-2}/4)] du,$$

which is the expectation of $F(U^{1/3}, X_{N-2}/4)$ with U independent of X_{N-2} , since $3u^2 du$ on $[0, 1]$ is the law of $U^{1/3}$.

(c) For $k \in (-\frac{5}{2}, 0]$ take $F(u, y) = y^{2k}$ and $k_0 = k$; for $k > 0$, y^{2k} is bounded on $(0, B_N]$ and $k_0 = 0$. For log-moments, $|\log y|^m \leq C_m(1 + y^{-1})$ on $(0, B_N]$, so $k_0 = -\frac{1}{2}$ applies; likewise for $(\log y)^m$ with integer m . Dividing by (a) gives the limits; cumulants are polynomials in moments. For the last statement let $N \geq 3$ and $k \leq -\frac{5}{2}$, so $a := k+2 \leq -\frac{1}{2}$. The density p_{N-2} is continuous and positive near a configuration with $\vartheta_1 = 0$ and the other angles distinct and away from 0, and there $|\Lambda_{N-2}(1)|^{2a} \geq c |2 \sin(\vartheta_1/2)|^{2a}$, which is not integrable near $\vartheta_1 = 0$; hence $\mathcal{M}_{N-2}(a) = \infty$. With $F = y^{2k} \geq 0$, Fatou's lemma applied to the (nonnegative) integrands $f_\varepsilon \rightarrow f_0$ gives $\liminf_\varepsilon \varepsilon^{-3} \mathbb{E} \sum_n \mathbf{1}\{s_n < \varepsilon\} (M_n/s_n^2)^{2k} \geq \frac{1}{6\pi} 4^{-2k} \mathcal{M}_{N-2}(a) = \infty$, while the denominator of $\langle \cdot \rangle_\varepsilon$ stays finite by (a). $\square$

Remark 3 (Position of the maximum). The maximum of $|\Lambda_N|$ over a small gap sits asymptotically at its midpoint: if t_n^* is the (smallest) maximiser on the n -th gap, then $\langle \mathbf{1}\{|(t^* - \theta_{(\cdot)})/s - \frac{1}{2}| > \eta\} \rangle_\varepsilon \rightarrow 0$ for every $\eta > 0$. Indeed, in the notation of the proof, with $t = sx$, $x \in [0, 1]$, the normalised profile is $x(1-x) \operatorname{sinc}(\frac{sx}{2}) \operatorname{sinc}(\frac{s(1-x)}{2}) |\Lambda_{N-2}(e^{isx})|$, which converges uniformly in x to $|\Lambda_{N-2}(1)|x(1-x)$; when $\Lambda_{N-2}(1) \neq 0$ the limit has the unique maximiser $\frac{1}{2}$, so the maximisers converge to $\frac{1}{2}$. Lemma C with the bounded integrand $\mathbf{1}\{|x^* - \frac{1}{2}| > \eta\}$, the bound $W_s \leq 4^{2(N-2)}$ and dominated convergence give the claim.

Remark 4 (Finite ε). All ε -dependence in (9) enters through functions of the unordered pair of endpoints $\{e^{\pm is/2}\}$ (after centring the arc at 1), hence through even functions of s . This suggests that the corrections to the limits in Theorem 1 are $O(\varepsilon^2)$; we have not proved it, but the Monte Carlo data of Section 7 are consistent with it.

4. THE EXACT LAW OF THE LIMITING HUMPS

Theorem 2. Let $n \geq 1$ and let $\xi_1, \dots, \xi_n$ be independent, with ξ_1 on the unit circle with law $\frac{1}{6}|1 - e^{i\alpha}|^4 \frac{d\alpha}{2\pi}$ and, for $2 \leq j \leq n$, ξ_j in the open unit disc $\mathbb{D}$ with density

$$c_j (1 - |z|^2)^{j-2} |1 - z|^4 \quad (\text{w.r.t. } d^2z), \quad c_j = \frac{j-1}{\pi} \cdot \frac{j(j+1)}{(j+2)(j+3)}.$$

Then $X_n \stackrel{d}{=} \prod_{j=1}^n |1 - \xi_j|$. Moreover:

- (a) $\tilde{\mathbb{E}}X_n^{2k} = \frac{\mathcal{M}_n(k+2)}{\mathcal{M}_n(2)} = \prod_{j=1}^n \frac{\Gamma(j+2)^2 \Gamma(j+4+2k)}{\Gamma(j+4) \Gamma(j+2+k)^2}$ for $k > -\frac{5}{2}$, and $\tilde{\mathbb{E}}X_n^{2k} = \infty$ for $k \leq -\frac{5}{2}$.
- (b) $t \mapsto \log \tilde{\mathbb{E}}e^{t \log X_n} = \log \mathcal{M}_n(2 + t/2) - \log \mathcal{M}_n(2)$ is analytic for $\operatorname{Re} t > -5$, and for $r \geq 1$

$$(10) \quad \kappa_r(\log X_n) = \sum_{j=1}^n \left[\psi^{(r-1)}(j+4) - 2^{1-r} \psi^{(r-1)}(j+2) \right];$$

$$\text{in particular } \kappa_1(\log X_n) = H_{n+2} + H_{n+3} - \frac{10}{3}.$$

Proof. By [8, Prop. 2.2], under the Haar measure of $U(n)$, $\det(1 - U) \stackrel{d}{=} \prod_{j=1}^n (1 - \gamma_j)$ with independent γ_j : γ_1 is uniform on the unit circle and, for $j \geq 2$, $\gamma_j = e^{i\omega_j} \sqrt{\beta_j}$ with ω_j uniform and $\beta_j \sim \text{Beta}(1, j-1)$ independent. (There the factors are written $1 + e^{i\omega_j} \sqrt{\beta_j}$, which has the same law since ω_j is uniform.) Passing to $z = \sqrt{\beta} e^{i\omega}$ ($d^2z = \frac{1}{2} d\beta d\omega$), γ_j has density $\frac{j-1}{\pi} (1 - |z|^2)^{j-2}$ on $\mathbb{D}$.

Tilting. Let μ_j be the law of γ_j and $m_j = \int |1 - z|^4 \mu_j(dz)$. For bounded measurable h , by independence,

$$\tilde{\mathbb{E}}h(X_n) = \frac{\mathbb{E}[\prod_j |1 - \gamma_j|^4 h(\prod_j |1 - \gamma_j|)]}{\prod_j m_j} = \int h\left(\prod_j |1 - z_j|\right) \prod_{j=1}^n \nu_j(dz_j),$$

with $\nu_j(dz) = m_j^{-1} |1 - z|^4 \mu_j(dz)$, because the Radon–Nikodym density $\prod_j |1 - z_j|^4 / \prod_j m_j$ is a product of functions of the separate coordinates; so the $\xi_j \sim \nu_j$ are independent. The constants: $m_1 = \int_0^{2\pi} 16 \sin^4(\alpha/2) \frac{d\alpha}{2\pi} = 6$; for $j \geq 2$, with $\beta = |z|^2$ and rotation invariance, $m_j = \mathbb{E}(1 + \beta)^2 + 2\mathbb{E}\beta = 1 + \frac{4}{j} + \frac{2}{j(j+1)} = \frac{(j+2)(j+3)}{j(j+1)}$, since $\mathbb{E}\beta = \frac{1}{j}$ and $\mathbb{E}\beta^2 = \frac{2}{j(j+1)}$. Hence

$c_j = \frac{j-1}{\pi m_j}$ as stated. (Equivalently, $m_j = \mathcal{M}_j(2)/\mathcal{M}_{j-1}(2)$.) The same decomposition is the case $\beta = 2$, $\delta = 2$ of the product formula for the circular Jacobi ensemble in terms of deformed Verblunsky coefficients [9, Thm. 3.2 and Prop. 4.3]: their γ_{n-j} has the law of our ξ_j , $1 \leq j \leq n$.

(a) By definition $\widetilde{\mathbb{E}}X_n^{2k} = \mathbb{E}_n|\Lambda_n(1)|^{4+2k}/\mathcal{M}_n(2)$, and (2) with $k+2$ in place of k gives the product; it is finite iff $k+2 > -\frac{1}{2}$ (see the proof of Theorem 1(c)).

(b) $\widetilde{\mathbb{E}}e^{t \log X_n} = \widetilde{\mathbb{E}}X_n^t = \mathcal{M}_n(2+t/2)/\mathcal{M}_n(2)$, and $\log \mathcal{M}_n(2+t/2) = \sum_j [\log \Gamma(j) + \log \Gamma(j+4+t) - 2 \log \Gamma(j+2+t/2)]$ is analytic for $\operatorname{Re} t > -5$. Its r -th derivative at $t = 0$ is $\sum_j [\psi^{(r-1)}(j+4) - 2 \cdot 2^{-r} \psi^{(r-1)}(j+2)]$, which is (10). For $r = 1$, $\psi(j+4) - \psi(j+2) = \frac{1}{j+2} + \frac{1}{j+3}$, and summing gives $(H_{n+2} - H_2) + (H_{n+3} - H_3) = H_{n+2} + H_{n+3} - \frac{10}{3}$. $\square$

Remark 5 (The tilt ladder in the unitary group). The same argument with the tilt $|\Lambda_n(1)|^{2b}$ gives

$$(11) \quad \kappa_r^{(b)}(n) = \sum_{j=1}^n \left[\psi^{(r-1)}(j+2b) - 2^{1-r} \psi^{(r-1)}(j+b) \right].$$

For $b = 0$ and $n = N$ this is $\log |\Lambda_N(1)|$ under Haar measure [19]. For $b = 1$ and $n = N-1$ it is exact at finite N for the derivative at an eigenvalue: $|\Lambda'_N(e^{i\theta_1})| = \prod_{j \geq 2} |e^{i\theta_1} - e^{i\theta_j}|$, and by (1) the other $N-1$ angles, given θ_1 , are $|\Lambda_{N-1}(e^{i\theta_1})|^2$ -tilted [17]. For $b = 2$ and $n = N-2$ it is the small-gap limit of Theorem 1.

5. LARGE N

Corollary 3. *Let $Y_N = (2\pi/N)^2 X_{N-2}/4$, the small-gap limit of $M_J/\tilde{s}_J^2$ (Theorem 1(b) with $\tilde{s} = Ns/2\pi$). As $N \rightarrow \infty$:*

(a) *for every fixed $k > -\frac{5}{2}$,*

$$\mathbb{E}Y_N^{2k} = \left(\frac{\pi}{N}\right)^{4k} \frac{\mathcal{M}_{N-2}(k+2)}{\mathcal{M}_{N-2}(2)} \sim \pi^{4k} \frac{12 G(k+3)^2}{G(2k+5)} N^{k^2}; \quad \text{e.g. } \mathbb{E}Y_N^2 \sim \frac{\pi^4 N}{720};$$

(b) $\mathbb{E} \log Y_N = H_N + H_{N+1} - \frac{10}{3} - \log 4 + 2 \log \frac{2\pi}{N} = 2 \log \pi + 2\gamma - \frac{10}{3} + O(N^{-1})$, and $2 \log \pi + 2\gamma - \frac{10}{3} = 0.110558\dots$;

(c) $\operatorname{Var} \log Y_N = \frac{1}{2} \log N + c_2 + O(N^{-1})$, with $c_2 = \frac{65}{18} + \frac{\gamma}{2} - \frac{\pi^2}{2} = -1.035083\dots$;

(d) $\kappa_3(\log Y_N) < 0$ for every $N \geq 3$, it decreases strictly with N , and

$$\begin{aligned} \kappa_3(\log Y_N) \rightarrow & -2[\zeta(2) - H_4^{(2)} - 4(\zeta(3) - H_4^{(3)})] \\ & + \frac{1}{2}[\zeta(2) - H_2^{(2)} - 2(\zeta(3) - H_2^{(3)})] = -0.127077\dots; \end{aligned}$$

more generally, for $r \geq 3$, $\kappa_r(\log Y_N) \rightarrow (-1)^r(r-1)! [A_4^{(r)} - 2^{1-r}A_2^{(r)}]$ with $A_a^{(r)} = \zeta(r-1) - H_a^{(r-1)} - a(\zeta(r) - H_a^{(r)})$;

(e) $(\log Y_N - \mathbb{E} \log Y_N) / \sqrt{\frac{1}{2} \log N} \Rightarrow \mathcal{N}(0, 1)$.

The typical close-pair hump in unfolded units, $M/\tilde{s}^2$, stays of order one, but its logarithm spreads like $\frac{1}{2} \log N$; the linear growth of $\mathbb{E}Y_N^2$ is carried by that log-normal tail. Note that the $O(N^{-1})$ terms in (c) are large at moderate N : the exact variance is 0.2987 at $N = 10$ and 0.6340 at $N = 24$, against 0.1162 and 0.5539 from $\frac{1}{2} \log N + c_2$. Finite- N comparisons should use (10).

We need two lemmas.

Lemma E (Barnes asymptotics). *For fixed real $a > -\frac{1}{2}$, as $n \rightarrow \infty$,*

$$\mathcal{M}_n(a) = \frac{G(1+a)^2}{G(1+2a)} \cdot \frac{G(n+1)G(n+1+2a)}{G(n+1+a)^2} \sim \frac{G(1+a)^2}{G(1+2a)} n^{a^2}.$$

Proof. The identity follows from $\prod_{j=1}^n \Gamma(j+c) = G(n+1+c)/G(1+c)$, a consequence of $G(z+1) = \Gamma(z)G(z)$. The classical expansion $\log G(z+1) = \frac{z^2}{2} \log z - \frac{3}{4}z^2 + \frac{z}{2} \log 2\pi - \frac{1}{12} \log z + \zeta'(-1) + O(z^{-1})$ [5] gives, for fixed c , $\log G(n+1+c) - \log G(n+1) = (cn + \frac{c^2}{2}) \log n - cn + \frac{c}{2} \log 2\pi + o(1)$. The combination with $c = 0, 2a, a, a$ and weights $1, 1, -1, -1$ leaves $(2a^2 - a^2) \log n = a^2 \log n$. This is the asymptotic of [19]. $\square$

Lemma F (Digamma sums). *For integers $a \geq 0, n \geq 1$,*

$$(12) \quad \sum_{j=1}^n \psi'(j+a) = H_{n+a} - H_a - a[\psi'(a+1) - \psi'(n+a+1)] + n\psi'(n+a+1),$$

$$(13) \quad \sum_{j=1}^{\infty} \psi^{(r-1)}(j+a) = (-1)^r (r-1)! A_a^{(r)} \quad (r \geq 3).$$

Proof. Write $\psi^{(r-1)}(x) = (-1)^r (r-1)! \sum_{m \geq 0} (x+m)^{-r}$ and put $l = j+a+m$. For fixed $l \geq a+1$ the number of $j \in [1, n]$ with $j \leq l-a$ is $\min(n, l-a)$. For $r = 2$ this gives $\sum_{l=a+1}^{a+n} (l-a)l^{-2} + n \sum_{l>a+n} l^{-2}$, which is (12) since $\sum_{l>c} l^{-2} = \psi'(c+1)$. For $r \geq 3$ and $n = \infty$ it gives $\sum_{l>a} (l-a)l^{-r} = A_a^{(r)}$. All rearrangements are of series of one sign. $\square$

Proof of Corollary 3. (a) By Theorem 1(c) and scaling, $\mathbb{E}Y_N^{2k} = (2\pi/N)^{4k} 4^{-2k} \tilde{\mathbb{E}}X_{N-2}^{2k}$. By Lemma E with $a = k+2$ and $a = 2$, and $G(3) = 1, G(5) = 12, \mathcal{M}_{N-2}(k+2)/\mathcal{M}_{N-2}(2) \sim 12G(k+3)^2G(2k+5)^{-1}N^{(k+2)^2-4}$; multiply by $(\pi/N)^{4k}$. For $k = 1$, $12G(4)^2/G(7) = 12 \cdot 4/34560 = 1/720$.

(b) $\kappa_1(\log X_{N-2}) = H_N + H_{N+1} - \frac{10}{3}$ by Theorem 2; add $\log \frac{1}{4} + 2 \log \frac{2\pi}{N}$ and use $H_m = \log m + \gamma + O(m^{-1})$.

(c) By (10) and (12) with $n = N-2, \kappa_2 = S_4 - \frac{1}{2}S_2$ where $S_a = \sum_{j \leq n} \psi'(j+a) = \log n + \gamma - H_a - a(\zeta(2) - H_a^{(2)}) + 1 + O(n^{-1})$, using $\psi'(a+1) = \zeta(2) - H_a^{(2)}, n\psi'(n+a+1) = 1 + O(n^{-1})$. With $H_4 = \frac{25}{12}, H_4^{(2)} = \frac{205}{144}, H_2 = \frac{3}{2}, H_2^{(2)} = \frac{5}{4}$: $\kappa_2 = \frac{1}{2} \log n + \frac{\gamma}{2} + \frac{1}{2} - \frac{25}{12} + \frac{3}{4} - \frac{5}{4} + \frac{205}{36} - 3\zeta(2) + O(n^{-1}) = \frac{1}{2} \log N + \frac{65}{18} + \frac{\gamma}{2} - \frac{\pi^2}{2} + O(N^{-1})$. Scaling does not change the variance.

(d) The summands $\tau_j = \psi''(j+4) - \frac{1}{4}\psi''(j+2)$ of $\kappa_3(\log X_n)$ are all negative. For $j = 1, \tau_1 = -2(\zeta(3) - H_4^{(3)}) + \frac{1}{2}(\zeta(3) - H_2^{(3)}) = -0.010261 \dots$. For $j \geq 2$ use $x^{-2} \leq |\psi''(x)| \leq x^{-2} + 2x^{-3}$ (compare $2 \sum_{m \geq 0} (x+m)^{-3}$ with $2 \int_0^\infty (x+t)^{-3} dt$): it suffices that $(j+4)^{-2} > \frac{1}{4}(j+4)(j+2)^{-3}$, i.e. $4^{1/3}(j+2) > j+4$, true for $j \geq 2$. Hence $\kappa_3(\log X_n)$ is negative and strictly decreasing in n , and its limit is $\sum_{j \geq 1} \tau_j$, evaluated by (13). The general r is the same computation; the series converge because $|\psi^{(r-1)}(x)| \leq (r-1)!x^{-r} + (r-2)!x^{1-r}$.

(e) By (c) and (d), $\kappa_2 \rightarrow \infty$ while κ_r stays bounded for each $r \geq 3$, so the cumulants of the standardised variable converge to those of $\mathcal{N}(0, 1)$; the moments converge, and since the normal law is determined by its moments, the standardised variable converges in law.

Replace $\sqrt{\kappa_2}$ by $\sqrt{\frac{1}{2} \log N}$ (Slutsky). $\square$

6. HOW MUCH DO SMALL GAPS CONTRIBUTE TO THE SECOND MOMENT?

Corollary 4. *For $N \geq 2$,*

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^{-7} \mathbb{E} \sum_{n=1}^N \mathbf{1}_{\{s_n < \varepsilon\}} M_n^2 = \frac{\mathcal{M}_{N-2}(3)}{224\pi} = \frac{(N-1)N^2(N+1)^3(N+2)^2(N+3)}{1\,935\,360\,\pi}.$$

Proof. $M_n^2 = \varepsilon^4 (s_n/\varepsilon)^4 (M_n/s_n^2)^2$. Apply (5) with the bounded continuous $F(u, y) = u^4 y^2$: the limit is $\frac{1}{2\pi} \int_0^1 u^6 du \cdot \frac{1}{16} \mathbb{E}_{N-2} |\Lambda_{N-2}(1)|^6 = \mathcal{M}_{N-2}(3)/(224\pi)$; then use (3). $\square$

In unfolded units, $\varepsilon = 2\pi\tilde{\varepsilon}/N$, the contribution of the gaps with $\tilde{s}_n < \tilde{\varepsilon}$ is

$$\mathbb{E} \sum_{\tilde{s}_n < \tilde{\varepsilon}} M_n^2 = \left(\frac{2\pi}{N}\right)^7 \frac{\mathcal{M}_{N-2}(3)}{224\pi} \tilde{\varepsilon}^7 (1 + o(1)), \quad \left(\frac{2\pi}{N}\right)^7 \frac{\mathcal{M}_{N-2}(3)}{224\pi} \sim \frac{\pi^6}{15120} N^2 = 0.0636 N^2.$$

The total second moment. To compare this with the sum over all gaps, let

$$V_N(\vartheta) := \prod_{j=1}^N 2 \sin \frac{\vartheta - \theta_j}{2},$$

a real, real-analytic function with $|V_N(\vartheta)| = |\Lambda_N(e^{i\vartheta})|$ and $V_N(\vartheta + 2\pi) = (-1)^N V_N(\vartheta)$. Winn [26, (1.1)–(1.2)] uses $Z_U(\vartheta) = \prod_j (1 - e^{i(\theta_j - \vartheta)})$ and $V_U(\vartheta) = \exp(iN \frac{\vartheta + \pi}{2} - \frac{i}{2} \sum_j \theta_j) Z_U(\vartheta)$. Since $1 - e^{i(\theta_j - \vartheta)} = 2i e^{i(\theta_j - \vartheta)/2} \sin \frac{\vartheta - \theta_j}{2}$, we have $Z_U(\vartheta) = i^N e^{\frac{i}{2}(\sum_j \theta_j - N\vartheta)} V_N(\vartheta)$, hence $V_U = e^{iN\pi/2} i^N V_N = (-1)^N V_N$. Following Winn, put $\tilde{F}_N(h, k) := \mathbb{E} |V_N(0)|^{2k-2h} |V'_N(0)|^{2h}$.

Proposition 5 (Total second moment). *Let $N \geq 2$. Almost surely*

$$\sum_{n=1}^N M_n^2 = \int_0^{2\pi} |V_N(\vartheta) V'_N(\vartheta)| d\vartheta,$$

hence $\mathbb{E} \sum_n M_n^2 = 2\pi \tilde{F}_N(\frac{1}{2}, 1)$, and as $N \rightarrow \infty$

$$\mathbb{E} \sum_{n=1}^N M_n^2 \sim \frac{e^2 - 5}{2} N^2 = 1.194528 \dots N^2.$$

Proof. Almost surely the θ_j are distinct. On the n -th gap V_N has no zero, and $(\log |V_N|)' = \frac{1}{2} \sum_j \cot \frac{\vartheta - \theta_j}{2}$ is strictly decreasing (its derivative is $-\frac{1}{4} \sum_j \sin^{-2} \frac{\vartheta - \theta_j}{2}$), with limits $+\infty$ and $-\infty$ at the left and right endpoints. Hence V_N^2 increases up to a unique point t_n^* and decreases after it, vanishing at both endpoints, so that $\int_{\text{gap}} |V_N V'_N| = \frac{1}{2} \int_{\text{gap}} |(V_N^2)'| = V_N(t_n^*)^2 = M_n^2$. The N gaps tile a period of the 2π -periodic function $|V_N V'_N|$, which gives the identity. Rotating the eigenangles by α replaces $V_N(\vartheta)$ by $V_N(\vartheta - \alpha)$, so $\mathbb{E} |V_N(\vartheta) V'_N(\vartheta)|$ does not depend on ϑ and $\mathbb{E} \sum_n M_n^2 = 2\pi \tilde{F}_N(\frac{1}{2}, 1)$. Finally, Winn proved $N^{-2} \tilde{F}_N(\frac{1}{2}, 1) \rightarrow (e^2 - 5)/(4\pi)$ [26, (1.11), proved in §7], a conjecture of Hughes. $\square$

The identity is the random-matrix counterpart of the reduction used by Conrey and Ghosh for Hardy's function (see [26, §1]), and the constants match: under the Riemann Hypothesis $\sum_{0 < \gamma \leq T} \max_{\gamma \leq t \leq \gamma+} |\zeta(\frac{1}{2} + it)|^2 \sim \frac{e^2 - 5}{4\pi} T (\log T)^2$ [11], i.e. a mean of $\frac{e^2 - 5}{2} \log T$ per gap, against

$\frac{e^2-5}{2}N$ per gap here. Winn’s Theorem 2.2 [26] also evaluates $\widetilde{F}_N(\frac{1}{2}, 1)$ at finite N ; for $k = 1$, $h = \frac{1}{2}$ it reduces to (Appendix A)

$$(14) \quad \mathbb{E} \sum_{n=1}^N M_n^2 = 2(N+1) \left[N - \sum_{p=2}^N \frac{(p-2)! 2^p N!}{(N-p)! (p+1)! p! N^{p-1}} \right]$$

(for $N = 2$ this is 10, as a direct computation confirms), so that $\mathbb{E} \sum_n M_n^2 / N^2 = 1.6055, 1.4381, 1.3457, 1.2949, 1.2696$ for $N = 6, 10, 16, 24, 32$. A Monte Carlo run (4×10^4 matrices for $N \leq 16$, 2×10^4 for $N = 24, 32$; maxima on the 63 interior points of a 65-point grid per gap with parabolic refinement) gives 1.604(5), 1.455(7), 1.339(8), 1.284(14), 1.258(14); the $N = 10$ value is 2.6σ high, and an independent run with 4×10^5 matrices gives 1.4394(20) there.

Gaps below $\tilde{\varepsilon}$ therefore carry a fraction $c_N \tilde{\varepsilon}^7 (1 + o(1))$ of the second moment, with $c_N = 0.140$ ($N = 6$) decreasing to 0.066 ($N = 32$) and $c_N \rightarrow \pi^6 / (7560(e^2 - 5)) = 0.0532$; this is about 10^{-6} for $\tilde{\varepsilon} = 0.2$. Small gaps are invisible in the mean of M_n^2 , which is the quantity controlled in [11, 18].

7. MONTE CARLO VERIFICATION

Method. Haar unitaries were generated by the QR decomposition of complex Ginibre matrices with the phases of the diagonal of R removed [22]. For every gap with unfolded length $\tilde{s} < 0.4$, $\log |\Lambda_N|$ was evaluated on the 39 interior points of a uniform 41-point grid across the gap and the maximum refined by a parabola through the largest grid value and its neighbours. Statistics are reported in the bands $\tilde{s} < \tilde{\varepsilon}$ for $\tilde{\varepsilon} \in \{0.4, 0.3, 0.2, 0.1, 0.05\}$, in angle units (M/s^2); the predictions are Theorem 1 with Theorem 2. The “rate” is $\#\{\text{gaps} < \varepsilon\} / (\text{matrices} \cdot \varepsilon^3)$, to be compared with $N^2(N^2 - 1)/(72\pi)$. No zeta data enter this section.

N	matrices	events	$\mathbb{E}[M/s^2]$			rate		
			Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$
6	$6 \cdot 10^5$	3 724	1.4784	1.4828(30)	1.4751(83)	5.570	5.392	5.405
10	$4 \cdot 10^5$	4 302	3.9279	3.935(11)	3.919(31)	43.77	42.96	43.36
16	$3 \cdot 10^5$	5 253	10.194	10.178(35)	10.059(98)	288.6	280.4	289.1
24	$2 \cdot 10^5$	5 229	23.903	24.13(10)	24.10(29)	1464	1419	1457

N	$\mathbb{E} \log(M/s^2)$			$\text{Var} \log(M/s^2)$			κ_3
	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm
6	0.3232	0.3276(22)	0.3229(62)	0.1486	0.1447	0.1448	−0.0439
10	1.2292	1.2323(29)	1.2330(81)	0.2987	0.2963	0.2834	−0.0688
16	2.1007	2.0998(34)	2.0822(95)	0.4691	0.4664	0.4781	−0.0872
24	2.8723	2.8827(39)	2.8804(110)	0.6340	0.6306	0.6326	−0.0990

TABLE 1. CUE Monte Carlo against Theorems 1 and 2 (angle units; statistical errors in the last digits; “events” is the number of gaps with $\tilde{s} < 0.1$). Files 200t_mc_N{6,10,16,24}.json.

Results. Table 1 shows agreement with the theorems at the level of the statistical errors: at $\tilde{s} < 0.1$ every deviation of the mean hump and of the log-mean is below 2σ and the counts agree within 2σ (Poisson errors). At $\tilde{s} < 0.2$ the small finite- ε bias discussed below is visible (largest in the log-mean at $N = 24$, $+0.010$). Further checks:

- *Independence.* The correlation of $\tilde{s}$ with $\log(M/s^2)$ falls with the band: 0.021, 0.013, 0.007, 0.000 at $\tilde{\varepsilon} = 0.4, 0.3, 0.2, 0.1$ ($N = 10$, file `200t_mc_N10.json`).
- *Lemma A.* The largest difference between $\log(M/s^2)$ and $\log(|\Lambda_{N-2}(\text{mid})|/4)$ over all events with $\tilde{s} < 0.1$ is at most 0.042 (all N ; files of Table 1).
- *Third cumulant.* Independent runs with 8×10^5 ($N = 10$) and 4×10^5 ($N = 16$) matrices give $\kappa_3 = -0.0639$ (8 526 events) and -0.0863 (7 021 events) at $\tilde{s} < 0.1$, against -0.0688 and -0.0872 (files `200t_mc3_N{10,16}.json`); the same runs give $\mathbb{E} \log(M/s^2) = 1.2357(59)$ and $2.1021(81)$.
- *Theorem 2 directly.* Sampling the ξ_j by rejection (4×10^5 products) gives, for $N = 10$, $\mathbb{E} \log(X/4) = 1.2286$ (1.2292), $\text{Var} = 0.2976$ (0.2987), $\kappa_3 = -0.0675$ (-0.0688), $\mathbb{E} X/4 = 3.924$ (3.928); at $N = 24$ the mean, variance and $\mathbb{E} X/4$ agree within 0.2%, and $\kappa_3 = -0.0963$ (-0.0990) (script `200t_carpim.py`, output `not8_ek/200t_carpim_seed7.out`).
- *Finite ε .* In a run with 3×10^6 matrices at $N = 10$ (file `not8_ek/200t_mc_N10_3e6_seed7.json`) the excess of $\mathbb{E}[M/s^2]$ over its limit is 0.0771(15), 0.0440(23), 0.0198(41) at $\tilde{\varepsilon} = 0.4, 0.3, 0.2$: the successive ratios 1.75 and 2.22 match $(4/3)^2 = 1.78$ and $(3/2)^2 = 2.25$, as expected from Remark 4. At $\tilde{\varepsilon} = 0.2$ the excesses of $\mathbb{E} \log$, $\text{Var} \log$ and κ_3 are $+0.0070(11)$, -0.005 and $+0.004$.

8. THE ZETA SIDE: A TILT LADDER AND A CONJECTURE

Observables. Let $L = \log(t/2\pi)$, and for consecutive zeros let $m_n = (\gamma_n + \gamma_{n+1})/2$, $L_n = \log(m_n/2\pi)$, $\tilde{\delta}_n = \delta_n L_n/2\pi$ and

$$Y_n := M_n/\tilde{\delta}_n^2.$$

For a close pair, $Z(t) \approx \frac{1}{2}Z''(m_n)(t - \gamma_n)(t - \gamma_{n+1})$ on the gap, so $M_n = \frac{1}{8}|Z''(m_n)|\delta_n^2(1 + O((\delta_n L)^2))$ and Y_n is the zeta analogue of $M/\tilde{s}^2$. The three rungs of the *tilt ladder* are

- $b = 0$: $x = \log |Z(t)|$ at random t (Keating–Snaith [19]);
- $b = 1$: $x = \log (|Z'(\gamma_n)| 2\pi/L)$ at the zeros (Hughes–Keating–O’Connell [17]);
- $b = 2$: $x = \log Y_n$ on close pairs, $\tilde{\delta}_n < \varepsilon$ (this note).

The random-matrix baseline of rung b is $\kappa_r^{\text{CUE},b}(N) := \kappa_r^{(b)}(N-b)$ of (11), evaluated at $N = L$ by analytic continuation in N : the sums in (11) are replaced by

$$\begin{aligned} \sum_{j=1}^n \psi^{(r-1)}(j+a) &= (n+a) \psi^{(r-1)}(n+a+1) - a \psi^{(r-1)}(a+1) \\ &\quad + (r-1) [\psi^{(r-2)}(n+a+1) - \psi^{(r-2)}(a+1)] \end{aligned}$$

for $r \geq 2$ (for $r = 1$ the bracket is replaced by $-n$), an identity for integer $n \geq 1$ (induction on n , using $\psi(x+1) = \psi(x) + 1/x$ and its derivatives) whose right side is analytic in $n > -a-1$ (scripts `200t_merdiven.py`, `200t_sabitler.py`).

Arithmetic cumulants. The Keating–Snaith conjecture [19] multiplies the moments of $|\zeta|$ by

$$a_k = \prod_p \left(1 - \frac{1}{p}\right)^{k^2} \sum_{m \geq 0} \left(\frac{\Gamma(m+k)}{m! \Gamma(k)}\right)^2 p^{-m},$$

and the Hughes–Keating–O’Connell conjecture for $\zeta'(\rho)$ [17] carries the same factor. Writing $\log a_{t/2} = \sum_r \kappa_r^{\text{arit}} t^r / r!$ and expanding ${}_2F_1(k, k; 1; x) = 1 + k^2 \text{Li}_2(x) + 2k^3 S_{1,2}(x) + O(k^4)$, with $S_{1,2}(x) = \sum_{m \geq 2} H_{m-1} x^m / m^2$, gives

$$\begin{aligned} \kappa_1^{\text{arit}} &= 0, & \kappa_2^{\text{arit}} &= \frac{1}{2} \sum_p \left[\log\left(1 - \frac{1}{p}\right) + \text{Li}_2\left(\frac{1}{p}\right) \right] = -0.088124, \\ \kappa_3^{\text{arit}} &= \frac{3}{2} \sum_p S_{1,2}\left(\frac{1}{p}\right) = +0.233653 \end{aligned}$$

(primes up to 3×10^6 ; the tails are below 10^{-8}).

Conjecture 6 (Close pairs; as formulated before the tests). *Let $T \rightarrow \infty$ and then $\varepsilon \rightarrow 0$, for zeros in $[T, 2T]$ with $\tilde{\delta}_n < \varepsilon$.*

- (i) $(\tilde{\delta}_n/\varepsilon, Y_n)$ are asymptotically independent, and $\tilde{\delta}_n/\varepsilon \xrightarrow{d} U^{1/3}$.
- (ii) $\mathbb{E}[Y_n^{2k} \mid \tilde{\delta}_n < \varepsilon] \sim a_k \mathbb{E}Y_L^{2k} \sim a_k \pi^{4k} 12 G(k+3)^2 G(2k+5)^{-1} L^{k^2}$, where Y_L is Y_N of Corollary 3 at $N = L$; in particular $\mathbb{E}[Y_n^2 \mid \tilde{\delta}_n < \varepsilon] \sim \pi^4 L / 720$ ($a_1 = 1$).
- (iii) $\kappa_r(\log Y_n \mid \tilde{\delta}_n < \varepsilon) = \kappa_r^{\text{CUE},2}(L) + \kappa_r^{\text{arit}} + o(1)$.

The heuristic is the hybrid Euler–Hadamard product of Gonek, Hughes and Keating [15], $\zeta \approx P_X Z_X$: conditioning on a close pair acts on Z_X as in Theorem 1, while P_X is not tilted at leading order, as in the treatment of $\zeta'(\rho)$ by Bui, Gonek and Milinovich [10]; the moments factor, and X cancels between $\mathbb{E}|P_X|^{2k} \sim a_k (e^\gamma \log X)^{k^2}$ and the tilted CUE of dimension $L/(e^\gamma \log X)$. Its weak points, recorded before the tests, are: (W1) the independence of P_X from the conditioning, whereas the prime waves modulate the local density of zeros at relative order $1/L$ (Landau’s formula [20, 16]) and imprint on the gaps, as measured in the companion notes [2, 3] — and at $b = 2$ the arithmetic term is not a small correction (κ_3 : CUE baseline -0.069 , arithmetic shift $+0.234$ at $L \approx 10$); (W2) finite L , where [1] found effective dimensions $N = L + c$; (W3) the order of limits and the transfer of finite- ε corrections; (W4) the passage from (ii) to (iii), which interchanges $L \rightarrow \infty$ with differentiation in k at $k = 0$.

Finite L . At $L \approx 11$ the form $a_k \times (\text{tilted CUE})$ is not the moment sequence of a probability law: $\log |a_{i\omega/2}|$ grows faster than ω^2 (each prime $p \lesssim \omega^2$ contributes), while the characteristic function of the tilted CUE decays only like $e^{-\omega^2 \log N/4}$. Its low cumulants are nevertheless defined. Genuine probability models are the hybrids $\text{hyb}(X)$: an independent random Euler product over $p \leq X$ times a tilted CUE of dimension $N_X = L/(e^\gamma \log X)$. Table 2 lists the predictions at $L = 10$. At $b = 2$ every arithmetic model predicts a positive κ_3 , whereas the pure tilted CUE gives -0.069 and a stationary Gaussian surface with the same power spectrum gives $\kappa_3 = \psi''(3/2)/8 = -0.1036$ and $\kappa_2 = \psi'(3/2)/4 = 0.2337$, independent of L (the two-point Kac–Rice density makes $|X''|/\sigma$ a χ_3 variable). The pre-test headline was therefore a sign: *close-pair log-humps of zeta are skewed to the right*, with skewness $\kappa_3/\kappa_2^{3/2} \approx +1.7$ at $L = 10$ under a_k .

b	CUE	a_k	hyb $X=2$	$X=3$	$X=5$	$X=7$
0	1.940/−2.393	1.852/−2.159	2.126/−2.233	2.080/−2.126	1.996/−2.042	1.977/−1.991
1	0.568/−0.194	0.480/+0.039	0.775/−0.039	0.790/+0.050	0.778/+0.103	0.803/+0.131
2	0.299/−0.069	0.211/+0.165	0.523/+0.083	0.585/+0.160	0.622/+0.197	0.674/+0.213

TABLE 2. Pre-test predictions κ_2/κ_3 of the log-observable on rung b at $L = 10$.

Gaussian surface: $b = 0$: $\pi^2/8 = 1.234$, $-7\zeta(3)/4 = -2.104$; $b = 1$: $\pi^2/24 = 0.411$, $-\zeta(3)/4 = -0.300$; $b = 2$: 0.234 , -0.104 .

9. PRE-REGISTERED TESTS ON THE RIEMANN ZEROS

Protocol. Each test was frozen — predictions, thresholds, analysis code — in a pre-registration file whose sha256 stamp was published before the measurement: 200-A on 26 September 2026, 17:12 (UTC+3), 200-B on 26 September 2026, 20:19, 200-C on 27 September 2026, 16:22. Before each measurement no distribution summary (moment, cumulant, quantile, histogram) of the tested observable had been computed on the data used; engine checks reported only differences between two engines. (The only earlier look at a rung-0 distribution in the project was a Kolmogorov–Smirnov test of $\log|Z(t)|$ against Selberg’s normal law at low height; Z' at the zeros and the humps of close pairs had never been computed.) Test 200-B was designed after 200-A was sealed, and 200-C after 200-B. Test 201 (phase lock) was frozen in the same way on 28 September 2026, 09:34, and committed and pushed to the (private) repository before its measurement; no phase statistic of the zeros had been computed before (only synthetic data).

Data and engine. Tests 200-A/B use A. M. Odlyzko’s first 2×10^6 zeros [24, 23] in three windows: W1, zeros 20868–198239 ($L \in [8, 10)$, $\bar{L} \approx 9.3$); W2, 198239–598742 ($[10, 11)$, 10.6); W3, 598742–2001052 ($[11, 12.1]$, 11.65). Test 200-C uses four LMFDB files of zeros computed by Platt [21, 25], 1.5×10^6 consecutive zeros each, at $\bar{L} = 14.19, 16.58, 18.88, 22.31$ (windows C1–C4). Z and Z' are computed by a vectorised Riemann–Siegel formula with the C_0 and C_1 remainder terms (anchored phases in C1–C4), checked against `mpmath` ($|\Delta Z| \leq 2.4 \times 10^{-7}$ in W1–W3, $\sim 10^{-9}$ in C1–C4). On rung $b = 0$ we use 10^6 uniform random t per window; on $b = 1$ all zeros (in C4, 10^6 random zeros); on $b = 2$ all pairs with $\tilde{\delta}_n < \varepsilon$, primary $\varepsilon = 0.2$ (1143, 2860, 10165 pairs in W1–W3; 11515 to 12143 in each of C1–C4), with M_n from a 129-point grid and parabolic refinement. Statistics are unbiased k -statistics k_2, k_3 ; errors are jackknife errors over 64 contiguous t -blocks (128 where a pre-registered block-sensitivity rule required it), rescaled by factors f calibrated on synthetic data processed by the same pipeline ($f = 0.82$ – 1.22 in 200-A, 0.94 – 1.13 in 200-B, 0.94 – 1.25 in 200-C).

Shifts. The *arithmetic shift* of rung b is $s_b := k_3 - \kappa_3^{\text{CUE},b}(\bar{L})$ (similarly for k_2). On $b = 2$ we also subtract the finite- ε correction $\Delta\kappa_r^{\text{CUE}}(\varepsilon)$ from a CUE Monte Carlo at the same ε and $N \approx \bar{L}$, and add 100% of it to the error, since its transfer to zeta is assumed, not tested; this defines $D_r := k_r - \kappa_r^{\text{CUE},2}(\bar{L}) - \Delta\kappa_r^{\text{CUE}}(\varepsilon)$, pooled over windows by inverse variance ($\bar{D}_r$).

Test 200-A: the lower rungs. A family of parameter-free models — CUE, a_k , hyb(X) for $X \in \{2, 3, 5, 7, 11\}$, Gaussian — was compared on the twelve numbers (k_2, k_3 on two rungs in three windows) with the full jackknife covariance plus a pre-declared theory margin (0.0188 for κ_2 , 0.0334 for κ_3).

Observation 1 (Lower rungs; pre-registered). *The arithmetic signature is present ($\chi_{\text{CUE}}^2 - \chi_{a_k}^2 = 287.1$, threshold 25), and the selected model is a_k , decisively (runner-up CUE at $\Delta\chi^2 = 287.1$, threshold 9), with goodness of fit $\chi_{\text{min}}^2/12 = 28.2/12 = 2.35$ (“good”). The two rungs separately select a_k ($\chi^2 = 6.1$ and 22.1). The hybrids and the Gaussian surface are rejected (χ^2 : CUE 315.2, a_k 28.2, hybrids 872–1243, Gaussian 3689.7; a continuous- X hybrid is best at $X = 2.99$ with $\chi^2 = 129.3$).*

Post hoc, the shifts tell a more detailed story (Table 3). The variance shift is nearly rung-independent and close to κ_2^{arit} : -0.0820 ± 0.0024 at $b = 0$ and -0.0726 ± 0.0008 at $b = 1$, against -0.0881 . The skewness shift is not: $+0.283 \pm 0.009$ at $b = 0$ and $+0.150 \pm 0.001$ at $b = 1$, against $+0.2337$, stable across the three windows. A ladder with the same arithmetic contribution on every rung therefore fails at the level of κ_3 at these heights. Before 200-B we registered, besides a_k , two extrapolations of the two lower rungs (not theories): geometric decay, $s_2 = s_1^2/s_0$, predicting $\bar{D}_3 = +0.0794 \pm 0.0028$; and linear decay, $s_2 = 2s_1 - s_0$, predicting $+0.017 \pm 0.0095$ (grouped with pure RMT, $\bar{D}_3 = 0$, as “extinguished”).

Test 200-B: close pairs.

Observation 2 (Close pairs; pre-registered, blind). *With 14 168 close pairs ($\tilde{\delta} < 0.2$, $L \approx 9$ –12):*

- (1) *The arithmetic signature is present on $b = 2$: $\bar{D}_3 = +0.0859 \pm 0.0030$ (29σ).*
- (2) *The sign of the skewness flips: the pooled corrected $\kappa_3 = +0.0143 \pm 0.0030$ (4.8σ). The skewness $\kappa_3/\kappa_2^{3/2}$ is about $+0.15$, not the $+1.7$ of the a_k form: the headline prediction held in sign, not in size.*
- (3) *The a_k model carried over from 200-A holds for the variance ($\bar{D}_2 = -0.0988$ against -0.0881 , tolerance 0.024) and fails for the third cumulant (difference -0.148 , tolerance 0.142): partial.*
- (4) *The shape of the ladder is geometric, decisively: $\chi^2 = 0.36$ (geometric), 24.0 (extinguished), 200.7 (constant a_k). The blind geometric prediction $+0.0794 \pm 0.0028$ differs from the measurement by 0.0065 (1.6σ).*

Secondary results (recorded, not decisive): $\bar{D}_3 = +0.0867 \pm 0.0066$ at $\varepsilon = 0.1$ and $+0.0828 \pm 0.0044$ at $\varepsilon = 0.3$; per window $+0.076(6)$, $+0.084(5)$, $+0.093(5)$, a mild rise with height ($\approx 2.5\sigma$). The variance shift on $b = 2$ is -0.0988 ± 0.0036 , far from the geometric and linear extrapolations (-0.064 , $\approx 9\sigma$). Theorem 1(b) has zeta counterparts: the Pearson correlation of $\tilde{\delta}_n$ and $\log Y_n$ is $+0.021$, -0.004 , -0.005 in W1–W3 (CUE Monte Carlo: $+0.005$ to $+0.008$), and the fraction of events with $\tilde{\delta}_n < \varepsilon/2$ is 0.124 – 0.129 (the $U^{1/3}$ law gives 0.125 , the CUE at finite ε 0.128).

rung	observable	κ_3 shift	ratio to previous rung	κ_2 shift
$b = 0$	$\log Z(t) $, random t	$+0.2828 \pm 0.0093$	—	-0.0820 ± 0.0024
$b = 1$	$\log Z'(\gamma_n) $	$+0.1499 \pm 0.0009$	0.530 ± 0.018	-0.0726 ± 0.0008
$b = 2$	$\log(M_n/\tilde{\delta}_n^2)$, $\tilde{\delta}_n < 0.2$	$+0.0859 \pm 0.0030$	0.573 ± 0.020	-0.0988 ± 0.0036
a_k	(every rung)	$+0.2337$	1	-0.0881

TABLE 3. Arithmetic shifts (observed minus tilted CUE at $N = \bar{L}$), pooled over W1–W3 ($L \approx 9$ –12).

Test 200-C: height. Is the decay a property of the ladder or of the height? Before 200-C, a post hoc fit of the A/B window values to $s_b(L) = a_k + c_b L^{-\nu}$ gave $\nu = 0.48 \pm 0.12$; three hypotheses were registered for the new windows: H_C (shifts constant at their A/B values), $H_{S\nu}$ (the fitted relaxation towards a_k) and H_{S1} ($\nu = 1$). The exponent is called γ in the pre-registration files.

Observation 3 (Height; pre-registered, blind). *On C1–C4 ($L = 14.2$ – 22.3), the class is $H_{S\nu}$, decisively ($\chi^2 = 10.3$ for 12 points; H_{S1} 30.1, H_C 166.2). Fitted on the new windows alone with asymptote a_k , the exponent is $\hat{\nu} = 0.749 \pm 0.170$ (4.4σ from zero: finite height). Over all seven windows with a free common asymptote, $\hat{A} = 0.277 \pm 0.041$ (within 1.0σ of $\kappa_3^{\text{arit}} = 0.2337$) and $\hat{\nu} = 0.447 \pm 0.144$.*

window	L	$b = 0$	$b = 1$	$b = 2$	s_1/s_0	s_2/s_1
W1–W3	9.3–11.7	+0.283	+0.150	+0.086	0.53	0.57
C1	14.19	+0.281(16)	+0.1633(13)	+0.1053(77)	0.58	0.64
C2	16.58	+0.289(20)	+0.1744(12)	+0.1183(60)	0.60	0.68
C3	18.88	+0.275(17)	+0.1787(15)	+0.1242(69)	0.65	0.70
C4	22.31	+0.283(19)	+0.1839(19)	+0.1466(91)	0.65	0.80

TABLE 4. Arithmetic shift of κ_3 by rung and height. Blind $H_{S\nu}$ predictions for C1–C4: $b = 1$: +0.159, +0.164, +0.169, +0.174; $b = 2$: +0.105, +0.114, +0.121, +0.130.

The observed $b = 1$ shifts lie 0.005–0.01 above the blind predictions (faster relaxation). The variance shifts do not depend on height: -0.082 to -0.094 ($b = 0$), -0.075 to -0.079 ($b = 1$), -0.095 to -0.115 ($b = 2$), around $\kappa_2^{\text{arit}} = -0.088$. The gap–hump correlation agrees with the CUE Monte Carlo ($|\rho| \leq 0.01$).

Test 201: phase lock at the conditioned points. The ladder concerns the size of $|Z|$ at conditioned points; a simpler question is where the conditioned points sit relative to the phase of a prime wave $\cos(t \log p)$. For single zeros the answer is Landau’s formula [20] (uniform version in [16]): for fixed $x > 1$, $\sum_{0 < \gamma \leq T} x^\rho = -\frac{T}{2\pi} \Lambda(x) + O(\log T)$, which for zeros on the critical line and a window at height t means

$$\mathbb{E}[\cos(m\gamma_n \log p)] \approx -\frac{\log p}{p^{m/2}L}, \quad \mathbb{E}[\sin(m\gamma_n \log p)] \approx 0.$$

Heuristically, the explicit formula modulates the local density of zeros, $n(t) \approx \frac{L}{2\pi} [1 - \frac{2}{L} \sum_{p,m} \frac{\log p}{p^{m/2}} \cos(mt \log p)]$, and the zeros avoid the crests of the prime waves; averaging $\cos(mt \log p)$ against $n(t)$ gives the same values. For the midpoints of close pairs we registered three classes for the ratio R_2 of their lock to the single-zero value: 1 (a pair behaves like one zero), 2 (the two densities multiply) and 4. The last is a heuristic, not a theorem: if the zeros are locally GUE at density $n(t)$, the density of pairs with gap δ at midpoint t is $n(t)^2 [1 - \text{sinc}^2(\pi \delta n(t))] \approx \frac{\pi^2}{3} \delta^2 n(t)^4$, and since the pairs are selected by $\tilde{\delta}_n = \delta_n L / 2\pi < \varepsilon$, i.e. with the mean rather than the local density, their midpoints are weighted by $n(t)^4$ and lock four times as strongly as single zeros; for larger ε the weight tends to $n(t)^2$, and R_2 towards 2.

On the seven windows W1–W3, C1–C4 and the primes $p \leq 13$ (ratios pooled over primes with the jackknife covariance, then over windows):

- Observation 4** (Phase lock; pre-registered). (1) *Single zeros follow Landau’s formula to about four decimals: the ratio of the measured lock to $-\log p/(\sqrt{p}L)$ is $R_1 = 1.0000 \pm 0.00004$ (per window from 1.003 ± 0.007 in $W1$ to 1.00003 ± 0.00009 in $C4$), also at the second harmonic (1.0000 ± 0.0001), with no drift in $\log L$; at $C4$, for $p = 2$, the mean of $\cos(\gamma_n \log 2)$ is -0.021975 against -0.021973 . At random t the same statistic is -0.010 to $+0.005$ (control).*
- (2) *Midpoints of close pairs ($\tilde{\delta}_n < 0.2$) lock $R_2 = 3.874 \pm 0.023$ times as strongly: class 4, with the classes 1 and 2 excluded. The ratio decreases with ε , $R_2 = 4.07 \pm 0.06$, 3.87 ± 0.02 , 3.74 ± 0.01 at $\varepsilon = 0.1, 0.2, 0.3$, as the heuristic expects; at the second harmonic it is 3.06 ± 0.05 , consistent with the second-order terms of the $n(t)^4$ weight (a synthetic process with that weight gives 2.90).*

For single zeros this verifies a theorem rather than discovers anything; a by-product is that the block-to-block fluctuations of these averages are 7–19 times smaller than for independent points, a sign of the rigidity of the zeros. The lock of close pairs has the size the local-GUE heuristic predicts, so there is no excess lock. It therefore does not by itself explain the decay of the κ_3 shift along the ladder: a translation of the measured lock into cumulant shifts through independent Euler factors was inconclusive (its sign depends on the harmonic cut-off), and the mechanism of the decay remains open.

Reading. The “halving at each conditioned zero” seen at $L \approx 11$ is a finite-height effect: as the height grows the ladder flattens, the ratios rising from 0.53/0.57 to 0.65/0.80, and the differences between rungs close like $L^{-\nu}$ with $\nu \approx 0.45$ –0.75. This is compatible with the asymptotic picture of Keating–Snaith and Hughes–Keating–O’Connell: the three rungs approach a common value, and the free asymptote 0.277 ± 0.041 is within 1σ of a_k . One point is open: the $b = 0$ shift stays flat at ≈ 0.28 over $L = 9$ –22, 0.05 above a_k , and these data cannot decide whether the common asymptote is a_k (0.234) or ≈ 0.28 . The geometric decay of 200-B and the relaxation law of 200-C are extrapolations, not theories; each passed a blind test on unseen data, but neither is derived. A phase bias of the primes at the conditioned points, which weakens like $1/L$, would produce a relaxation of this kind; Test 201 shows that such a bias is present with the predicted size (Landau’s formula for single zeros, close to four times as much for close pairs), but we have not been able to connect it quantitatively with the κ_3 shifts. As in [4], where a post hoc pattern $\eta_p \approx p^{-1/3}$ of transfer factors turned out to be a finite-height effect, a pattern seen at one height (≈ 0.55 per rung) was a coincidence of height.

10. OPEN PROBLEMS

- (1) *Finite- ε corrections.* Prove that the corrections in Theorem 1 are $O(\varepsilon^2)$ and compute the leading coefficient.
- (2) *Close pairs of zeta.* Prove Conjecture 6(ii) for small k under a hypothesis, in the manner of [10] for $\zeta'(\rho)$, or from the ratios conjecture.
- (3) *The skewness ladder.* Derive the rung dependence of the κ_3 shift at finite height and its relaxation rate $L^{-\nu}$; decide whether the $b = 0$ plateau at ≈ 0.28 bends down to a_k , and whether the phase lock of Test 201, which has the predicted size, plays a role in it.
- (4) *Phase lock of close pairs.* Derive the factor 4 of Observation 4 and its ε -dependence beyond the local-GUE heuristic.

- (5) *All gaps.* Theorem 1 is the small-gap end of the open problem of [1]: the mean of M_n^2 conditioned on a gap of any size.

APPENDIX A. WINN'S FORMULA FOR $h = \frac{1}{2}$, $k = 1$

We specialise [26, Thm. 2.2, eq. (2.9)] to $h = \frac{1}{2}$, $k = 1$, in Winn's notation. For a partition λ and $b \in \mathbb{R}$, $[b]_\lambda = \prod_{i=1}^{\ell(\lambda)} (b - i + 1)_{\lambda_i}$ with the rising factorial $(x)_m = x(x+1) \cdots (x+m-1)$, h_λ is the product of the hook lengths of λ , and

$$C_N(p, k) = (-2)^p \sum_{\lambda \vdash kp} \frac{[k]_\lambda [-N]_\lambda}{[2k]_\lambda h_\lambda^2},$$

the sum running over partitions of p into at most k parts [26, (2.1)–(2.5)]. For half-integer h and $k \in \mathbb{N}$, [26, (2.9)] reads

$$\begin{aligned} \tilde{F}_N(h, k) = \frac{2(-1)^{h+1/2}}{2^{2h}\pi} \tilde{F}_N(0, k) & \left\{ \sum_{p=1}^{2h} \sum_{\ell=1}^p \binom{2h}{p-\ell} \frac{(-1)^\ell}{\ell} (-N)^{2h-p} p! C_N(p, k) \right. \\ & \left. + \sum_{p=2h+1}^{kN} \frac{(2h)!(p-2h-1)!}{N^{p-2h}} C_N(p, k) \right\}. \end{aligned}$$

Step 1: the ingredients. For $h = \frac{1}{2}$, $k = 1$: the prefactor is $2(-1)^1/(2\pi) = -1/\pi$; $\tilde{F}_N(0, 1) = \mathcal{M}_N(1) = \prod_{j=1}^N \frac{\Gamma(j)\Gamma(j+2)}{\Gamma(j+1)^2} = \prod_{j=1}^N \frac{j+1}{j} = N+1$ by (2); the first double sum has the single term $p = \ell = 1$, equal to $\binom{1}{0} \frac{(-1)^1}{1} (-N)^0 1! C_N(1, 1) = -C_N(1, 1)$; and the second sum runs over $2 \leq p \leq N$ with weight $(p-2)!/N^{p-1}$.

Step 2: the partition sum. For $k = 1$ only the one-part partition $\lambda = (p)$ contributes. Its hook lengths are $p, p-1, \dots, 1$, so $h_{(p)} = p!$, and $[b]_{(p)} = (b)_p$, so that $[1]_{(p)} = p!$, $[2]_{(p)} = (p+1)!$ and $[-N]_{(p)} = (-N)(-N+1) \cdots (-N+p-1) = (-1)^p N!/(N-p)!$ for $p \leq N$. Hence

$$C_N(p, 1) = (-2)^p \frac{p! (-1)^p N!/(N-p)!}{(p+1)! (p!)^2} = \frac{2^p N!}{(N-p)! (p+1)! p!}, \quad C_N(1, 1) = N.$$

Step 3: assembly. Collecting,

$$\tilde{F}_N(\tfrac{1}{2}, 1) = \frac{N+1}{\pi} \left[N - \sum_{p=2}^N \frac{(p-2)! 2^p N!}{(N-p)! (p+1)! p! N^{p-1}} \right],$$

and $\mathbb{E} \sum_n M_n^2 = 2\pi \tilde{F}_N(\frac{1}{2}, 1)$ (Proposition 5) gives (14). Each term is rational, so $\mathbb{E} \sum_n M_n^2 \in \mathbb{Q}$; for $N = 2, 3, 4, 6$ it equals $10, \frac{496}{27}, \frac{233}{8}, \frac{42134}{729}$.

Checks. (i) For $N = 2$ the bracket is $2 - \frac{1}{3}$, so the formula gives 10. Directly: let $\phi \in (0, 2\pi)$ be $\theta_2 - \theta_1$ modulo 2π ; the two gaps are ϕ and $2\pi - \phi$, and by the proof of Lemma A (with $N - 2 = 0$ other angles) a gap of length s has hump $4 \sin^2 \frac{s}{4}$, attained at its midpoint. The humps are thus $4 \sin^2 \frac{\phi}{4}$ and $4 \cos^2 \frac{\phi}{4}$, so $\sum_n M_n^2 = 16(\sin^4 \frac{\phi}{4} + \cos^4 \frac{\phi}{4}) = 16 - 8 \sin^2 \frac{\phi}{2}$, and ϕ has density $\sin^2(\phi/2)/\pi$ on $(0, 2\pi)$ by (1), so $\mathbb{E} \sin^2 \frac{\phi}{2} = \frac{3}{4}$ and $\mathbb{E} \sum_n M_n^2 = 10$. (ii) As $N \rightarrow \infty$, $N!/((N-p)!N^p) \rightarrow 1$ for each p and the terms are dominated by $2^p(p-2)!/((p+1)!p!)$, so $N^{-2} \tilde{F}_N(\frac{1}{2}, 1) \rightarrow \frac{1}{\pi} [1 - \sum_{p \geq 2} \frac{2^p(p-2)!}{p!(p+1)!}] = \frac{e^2-5}{4\pi}$, which is [26, (7.8)].

REPRODUCIBILITY

All code, data products, pre-registration files (with sha256 stamps and dates) and the research log, including the failed predictions, are available in the repository <https://github.com/azizran/riemann-zero-statistics> (directory `qm_riemann/`), archived at Zenodo, doi:10.5281/zenodo.22942475. Paths below are relative to `qm_riemann/200_configs/`, whose subdirectory `not8_ek/` holds the runs added for this note.

- Table 1, independence and Lemma A checks: `200t_mc_N*.json` ($N = 6, 10, 16, 24$; script `200t_mc.py`, predictions `200t_tahmin.py`); third cumulant: `200t_mc3_N*.json` ($N = 10, 16$); finite ε : `not8_ek/200t_mc_N10_3e6_seed7.json` (arguments 10 3000000 7); Theorem 2 by sampling and κ_2^{arit} : `200t_carpim.py`, output `not8_ek/200t_carpim_seed7.out`.
- Proposition 5: Monte Carlo `not8_ek/m2total.py`, output `not8_ek/m2total_seed11.out`; independent run `not8_ek/m2total_N10_4e5_seed2026.json`; exact values from Winn's formula `not8_ek/m2_exact_winn.json` (these two JSON files contain their generating scripts).
- Constants, ladder baselines and Table 2: `200t_sabitler.py`, `200t_merdiven.py`, `200t_hibrit.py`, `200t_aritmetik_kesin.py`.
- Tests 200-A/B/C: pre-registrations `ONKAYIT_200A/B/C.json` (sha256 prefixes `f0f0ce57`, `b37dc244`, `391257a4`; commits `aa861b5`, `85f0489`, `567af03`), verdicts `HUKUM_200A/B/C.json` (which also hold the CUE Monte Carlo comparisons of Section 9), synthetic calibrations and finite- ε CUE corrections `M6_200A.json`, `M6b_200B.json`, `M6c_200C.json`. Test 201: directory `qm_riemann/201_configs/`, pre-registration `ONKAYIT_201.json` (sha256 prefix `ecc7cf75`, commit `16e7c3b`), verdict `HUKUM_201.json`.

The LMFDB zero files are not redistributed; their sha256 stamps are in `ONKAYIT_200C.json`.

ACKNOWLEDGEMENTS

The question that started this note was asked by a reader; we thank them for it. Computations, derivation audits, literature search and drafting were done with the help of an AI assistant (Claude, Anthropic); derivations and pre-registrations were audited by an independent model instance before predictions were frozen. Errors are the author's.

REFERENCES

- [1] U. Sezen, *The joint law of zero gaps and local maxima of Hardy's Z-function*, preprint (2026); archived at Zenodo, doi:10.5281/zenodo.22942475.
- [2] U. Sezen, *The prime-wave anatomy of the gap-amplitude law for Riemann zeros*, preprint (2026); archived at Zenodo, doi:10.5281/zenodo.22942475.
- [3] U. Sezen, *A response theory for the Riemann zero gas*, preprint (2026); archived at Zenodo, doi:10.5281/zenodo.22942475.
- [4] U. Sezen, *Arithmetic satellites of the Bragg comb of the Riemann zero lattice: residue classes, characters, and a blind test of the Bogomolny-Keating mirror law*, preprint (2026); archived at Zenodo, doi:10.5281/zenodo.22942475.
- [5] E. W. Barnes, *The theory of the G-function*, Quart. J. Pure Appl. Math. **31** (1900), 264–314.
- [6] G. Ben Arous, P. Bourgade, *Extreme gaps between eigenvalues of random matrices*, Ann. Probab. **41** (2013), no. 4, 2648–2681, doi:10.1214/11-AOP710.
- [7] P. Bourgade, *Extreme gaps between eigenvalues of Wigner matrices*, J. Eur. Math. Soc. **24** (2022), no. 8, 2823–2873, doi:10.4171/JEMS/1141.

- [8] P. Bourgade, C. P. Hughes, A. Nikeghbali, M. Yor, *The characteristic polynomial of a random unitary matrix: a probabilistic approach*, Duke Math. J. **145** (2008), no. 1, 45–69, doi:10.1215/00127094-2008-046; arXiv:0706.0333 (statement numbers cited here are those of the arXiv version).
- [9] P. Bourgade, A. Nikeghbali, A. Rouault, *Circular Jacobi ensembles and deformed Verblunsky coefficients*, Int. Math. Res. Not. IMRN **2009**, no. 23, 4357–4394, doi:10.1093/imrn/rnp092; arXiv:0804.4512 (statement numbers cited here are those of the arXiv version).
- [10] H. M. Bui, S. M. Gonek, M. B. Milinovich, *A hybrid Euler–Hadamard product and moments of $\zeta'(\rho)$* , Forum Math. **27** (2015), 1799–1828, arXiv:1302.5032.
- [11] J. B. Conrey, A. Ghosh, *A mean value theorem for the Riemann zeta-function at its relative extrema on the critical line*, J. London Math. Soc. (2) **32** (1985), 193–202.
- [12] R. Feng, G. Tian, D. Wei, *Small gaps of GOE*, Geom. Funct. Anal. **29** (2019), no. 6, 1794–1827, doi:10.1007/s00039-019-00520-5.
- [13] R. Feng, D. Wei, *Small gaps of circular β -ensemble*, Ann. Probab. **49** (2021), no. 2, 997–1032, doi:10.1214/20-AOP1468.
- [14] P. J. Forrester, *Log-Gases and Random Matrices*, London Math. Soc. Monographs Ser. **34**, Princeton University Press, Princeton, NJ, 2010.
- [15] S. M. Gonek, C. P. Hughes, J. P. Keating, *A hybrid Euler–Hadamard product for the Riemann zeta function*, Duke Math. J. **136** (2007), 507–549.
- [16] S. M. Gonek, *An explicit formula of Landau and its applications to the theory of the zeta-function*, in: M. Knopp et al. (eds.), *A Tribute to Emil Grosswald: Number Theory and Related Analysis*, Contemp. Math. **143**, Amer. Math. Soc., Providence, RI, 1993, 395–413.
- [17] C. P. Hughes, J. P. Keating, N. O’Connell, *Random matrix theory and the derivative of the Riemann zeta function*, Proc. R. Soc. Lond. A **456** (2000), 2611–2627.
- [18] C. Hughes, S. Lugmayer, A. Pearce-Crump, *The second moment of the Riemann zeta function at its local extrema*, J. London Math. Soc. (2) **112** (2025), e70250, arXiv:2411.05573.
- [19] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Comm. Math. Phys. **214** (2000), 57–89.
- [20] E. Landau, *Über die Nullstellen der Zetafunktion*, Math. Ann. **71** (1912), 548–564, doi:10.1007/BF01456808.
- [21] The LMFDB Collaboration, *The L-functions and modular forms database*, <https://www.lmfdb.org>, 2026, [Online; accessed 27 September 2026].
- [22] F. Mezzadri, *How to generate random matrices from the classical compact groups*, Notices Amer. Math. Soc. **54** (2007), 592–604, arXiv:math-ph/0609050.
- [23] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), 273–308.
- [24] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, https://www-users.cse.umn.edu/~odlyzko/zeta_tables/.
- [25] D. J. Platt, *Computing $\pi(x)$ analytically*, Math. Comp. **84** (2015), no. 293, 1521–1535, doi:10.1090/S0025-5718-2014-02884-6.
- [26] B. Winn, *Derivative moments for characteristic polynomials from the CUE*, Comm. Math. Phys. **315** (2012), no. 2, 531–562, doi:10.1007/s00220-012-1512-1; arXiv:1109.0227.

INDEPENDENT RESEARCHER, ISTANBUL, TURKEY

Email address: usezen@aof.anadolu.edu.tr